Feature-Composed Item Embeddings

SAMPLE v3 (thesis-write refinement, 2026-08-25) | discussion material, not thesis prose. v2 + new title (embeddings, not factors, "The Model:" dropped), pivot sentence aligned, requirements opening added.

The previous section committed us to grounding on the item side. Before presenting the design, we fix what it must deliver. The design must

- relate attributes to prototypes in an intrinsic and interpretable way, readable from the trained parameters rather than recovered after training,
- keep the ProtoMF paradigm, so that whatever changes can be attributed to the one modification,
- make the attributes available as a source of predictive power, not only as labels for interpretation.

We introduce our approach as feature-composed I-ProtoMF, in short fI-ProtoMF. It differs from I-ProtoMF [2] in exactly one place. The host draws the embedding $\mathbf{t}$ of every item from a lookup table with one free learned row per item. fI-ProtoMF replaces this table by a composition in the manner of LightFM [1],

$$\mathbf{t} = \sum_{f \in \mathcal{F}_t} \mathbf{e}_f \ \big(+\, \mathbf{e}_{\mathrm{ID}(t)}\big), \tag{1}$$

where $\mathcal{F}_t$ holds the feature values recorded for the item, each $\mathbf{e}_f \in \mathbb{R}^d$ is a learned vector, and $\mathbf{e}_{\mathrm{ID}(t)}$ is an optional per-item row discussed below. Everything downstream remains the host. The L^t item prototypes $\mathbf{p}_l^t$ stay free learned vectors, the item representation $\mathbf{t}^* \in \mathbb{R}^{L^t}$ is still the vector of similarities $\mathrm{sim}(\mathbf{t}, \mathbf{p}_l^t)$, the user is still a free vector $\mathbf{u} \in \mathbb{R}^{L^t}$, and the score I-score$(u,t) = \mathbf{u}^\top \mathbf{t}^*$ is trained with the host's loss and regularizers (Section 2.3). The model name keeps the f of feature rather than attribute because the composition sums whatever encoded rows it is given, the ID row being the first example, and choosing attribute-derived rows is what makes the resulting model interpretable.

The composition turns attribute values into learned words. Every distinct value that appears in the catalogue, the colour group black or the garment group jersey, owns exactly one row $\mathbf{e}_f$, shared by all items that carry it. An item is then spelled from these words. A plain black t-shirt sums the rows of its recorded values for department, product type, section, colour group, and pattern, and a second shirt that differs only in colour shares four of its five summands. The host's item table, one independent row per item, becomes a small vocabulary of attribute rows from which all item embeddings are assembled. The rows are trained by the collaborative signal like any other parameter, but each row receives the gradients of every interaction with every item that carries its value, so what a row comes to encode is the collaborative meaning of one attribute value across the catalogue.

This is the grounding announced in the previous section, stated precisely. What the composition constrains directly is the item embedding. Every $\mathbf{t}$ is a sum of rows that carry meaning on their own. The prototypes are not constrained at all. They remain free vectors, but they anchor a space that is now populated by attribute words, and a prototype can therefore be read by comparing it to those words directly in parameter space, with no detour through training data. Grounding the prototypes is in this sense shorthand for grounding the space they anchor, and the read-outs this buys are developed in Section 5.4.

What the composition gains can be stated as one identity. For any prototype $\mathbf{p}$, the inner product with an item embedding splits over the summands of Equation 1,

$$\langle \mathbf{t}, \mathbf{p} \rangle \;=\; \sum_{f \in \mathcal{F}_t} \langle \mathbf{e}_f, \mathbf{p} \rangle \;\; \big(+\, \langle \mathbf{e}_{\mathrm{ID}(t)}, \mathbf{p} \rangle\big), \tag{2}$$

and dividing by the norms turns the item cosine into a norm-weighted sum of attribute cosines,

$$\cos(\mathbf{t}, \mathbf{p}) \;=\; \sum_{f \in \mathcal{F}_t} \frac{\|\mathbf{e}_f\|}{\|\mathbf{t}\|} \, \cos(\mathbf{e}_f, \mathbf{p}), \tag{3}$$

where the ID term, when present, joins the sum like any other row. The host's shifted similarity adds a constant and changes nothing here. The host interprets a prototype by ranking all items by $\cos(\mathbf{t}, \mathbf{p})$ and reading the metadata of the top of the list, so it shows aggregates of exactly the terms that Equation 3 names individually. An item is near a prototype because its sum aligns, and the list alone cannot say whether one attribute carries the alignment, several share it, or the ID row does the work. Ranking the rows $\mathbf{e}_f$ by $\cos(\mathbf{e}_f, \mathbf{p})$ reads the terms themselves, one meaning at a time, with no detour through the catalogue.

The same equations show what is lost. Every item embedding now lies in the span of the vocabulary rows, so the item table's capacity drops from one free direction per item to at most one per attribute value, and two items that record the same attribute values receive the same embedding, the same representation, and the same score for every user unless the ID row separates them. Equation 3 also carries a caveat for the read-out itself. Its weights $\|\mathbf{e}_f\|/\|\mathbf{t}\|$ are set by the row norms, so a row with a large norm dominates every item cosine it enters, and the nearest items of a prototype and its nearest attribute rows need not agree. A prototype may also align with a direction between words that no single row captures, in which case the attribute ranking is flat while the item ranking still concentrates. The two read-outs are therefore complements rather than substitutes, and Section 5.4 uses both.

The optional ID row. The row $\mathbf{e}_{\mathrm{ID}(t)}$ in Equation 1 is a free per-item vector added to the sum by configuration. It restores per-item capacity, letting an item deviate from what its attribute values alone would predict, at the price of an ungrounded summand in every embedding. It also makes the host a special case. With $\mathcal{F}_t$ empty and the ID row kept, the model reduces to I-ProtoMF exactly, and our implementation reproduces the host bit-identically in this configuration. The chapter therefore evaluates two arms against the host. fI-ProtoMF uses features and the ID row, and the noid arm uses features alone. What the ID row contributes, and what its ungrounded share costs, is measured in Sections 5.4 and 5.5.

References

[1] M. Kula. Metadata embeddings for user and item cold-start recommendations. In *Proceedings of the 2nd Workshop on New Trends on Content-Based Recommender Systems (CBRecSys) at RecSys*, 2015.

[2] A. B. Melchiorre, N. Rekabsaz, C. Ganhör, and M. Schedl. Protomf: Prototype-based matrix factorization for effective and explainable recommendations. In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*, pages 246–256, 2022.